Homework 2- Due Date Feb 23, 11:59 P.M.

Problem 1. A box contains b balls, out of which exactly r are red. We select k of the balls at random, without replacement. What is the probability that m of the selected balls are red?

Problem 2. Out of eight women and eight men, we form a committee consisting of six different people. Assuming that each committee of size six is equally likely, find the probabilities of the following events:

(a) The committee consists of three men and three women.

(b) The committee has at least one woman.

(c) For the remainder of the problem, assume that Emma and Jack are among the sixteen people being considered. Both Emma and Jack are members of the committee.

Problem 3. We want to arrange 3 novels, 2 mathematics books, and 1 chemistry book on a bookshelf

(b) What is the probability that the mathematics books are together and the novels are together?

(c) What is the probability that the novels are together, but the other books can be arranged in any order?

Problem 4. Suppose $P(A) = 0.4$, $P(B') = 0.3$, and $P(A \cap B') = 0.1$.
(a) Find $P(A \cup B)$.
(b) Find $P(B'|A)$.
(c) Find $P(B|A')$.

Problem 5. We have an infinite collection of biased coins, indexed by the positive integers. Coin n has probability $\frac{2}{3^n}$ of being selected. A flip of coin n results in Tail with probability $\frac{1}{2^n}$. We select a coin and flip it. What is the probability that the result is Tail?

The probability that the result is Tail is:

Problem 6. It is estimated that 50% of emails are spam emails. Some software has been applied to filter these spam emails before they reach your inbox. A certain brand of software claims that it can detect 99% of spam emails, and the probability for a false positive (a non-spam email detected as spam) is 5%.

Now if an email is detected as spam, then what is the probability that it is in fact a non-spam email?